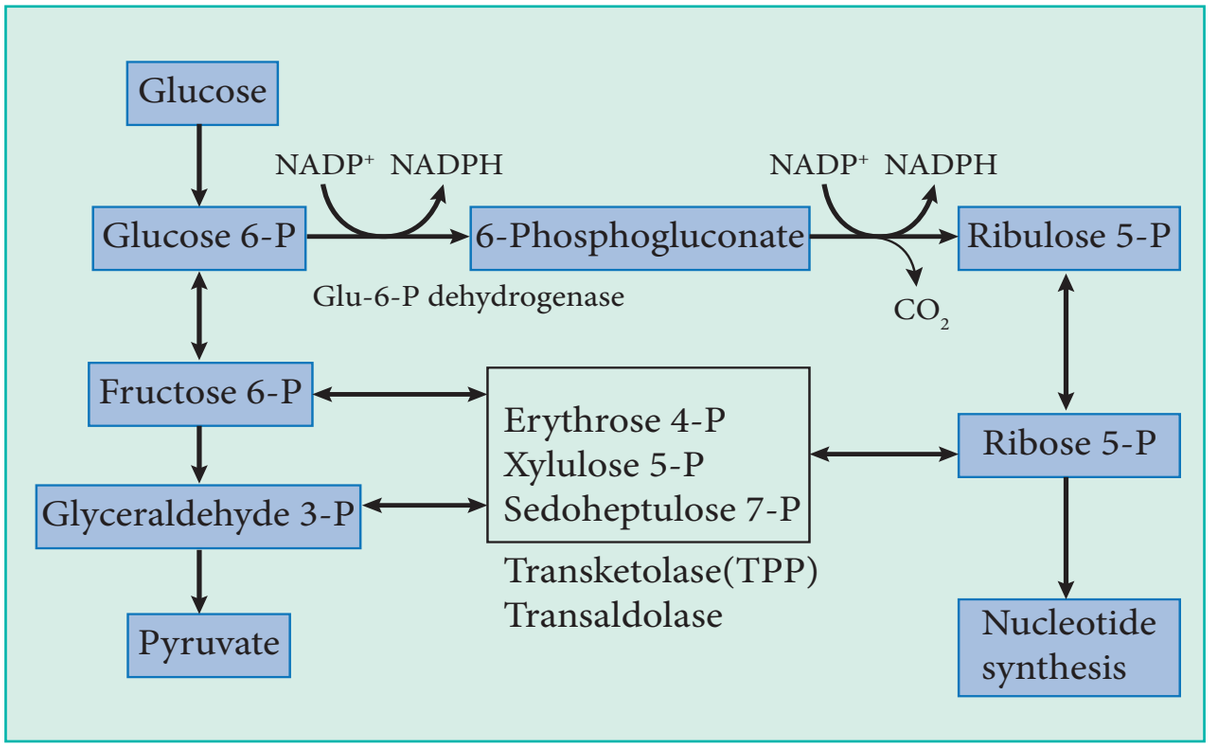
Step 2: Transketolase transfers a C2 unit from one xylulose-5-phosphate to the ribose-5-phosphate, yielding glyceraldehyde-3-phosphate and the C7 sugar sedoheptulose-7-phosphate.

Step 3: Transaldolase transfers a C3 unit from sedoheptulose-7-phosphate back to glyceraldehyde-3-phosphate, which yields fructose-6-phosphate and the C4 sugar erythrose-4-phosphate.

Step 4: Transketolase transfers a C2 unit from the second molecule of xylulose-5-phosphate to erythrose-4-phosphate. This yields a second molecule of fructose-6-phosphate and again glyceraldehyde-3-phosphate



3.6. GLYCOGEN METABOLISM

Glycogen is the major storage form of carbohydrates in animals, present in all cells, but found mostly in liver and muscle, where it occurs as cytoplasmic granules. It is a highly branched form of amylopectin, with branch points occurring every 8 to 14 glucose residues. Glycogen's highly branched structure, which has many non-reducing ends, permits the rapid mobilization of glucose in times of metabolic need.

3.6.1. Glycogenesis

The biosynthesis of glycogen from glucose is called Glycogenesis. Glycogenesis takes place when blood glucose levels are sufficiently high to allow excess glucose to be stored.

Glycogenesis is stimulated by the hormone insulin. Insulin facilitates the uptake of glucose into muscle cells, though it is not required for the transport of glucose into liver cells. However,

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